

Runguo Li

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Research Interests

ML Systems for large-scale models: inference engines and expert streaming for MoE models, distributed training and rollout infrastructure (ZeRO-3, FSDP, Hybrid Engine), RL post-training systems, and GPU compiler and kernel-level correctness. I work between the machine and the model — how weights are read from storage, how collectives and thread teams are sized, and whether a resource plan and the runtime agree about what will actually be built — while keeping a research background in LLM reasoning, multimodal learning, and agent safety.

Education

University of Illinois Urbana-Champaign (UIUC)

Aug 2026–Jul 2028

M.S. in Information Science

Urbana-Champaign, IL, USA

Shanghai University of Finance and Economics (SUFU)

Sep 2022–Jun 2026

B.S. in Data Analytics

Shanghai, China

Open-Source Contributions (ML Systems)

Merged fixes to a pure-C MoE inference engine, vLLM, SGLang, DeepSpeed, Triton, verl, LMCache, vLLM-Omni, Apache TVM, JAX, and FlashAttention.

vLLM — LLM serving engine

[Merged pull requests](#)

fixed the `int4_per_token_head` KV-cache quantization path on non-power-of-two head sizes, where the Hadamard-rotation write assumed a power-of-two dimension and aborted inside the engine-core process with an opaque error (PR #56198)

SGLang — LLM serving

[Merged pull requests](#)

deferred `deep_ep` import-time failures to first use, so a host check raising beyond `ImportError` (a second `libccl`, a missing `CUDA` home) no longer kills every server process at import, including dense models that never request `DeepEP`; added a CPU regression test that runs each case in a fresh interpreter (PR #40671)

colibri — pure-C MoE inference engine

[Merged pull requests](#)

multi-drive expert streaming for the 510 GB DeepSeek-V4.1 container, physical-core OpenMP team sizing in four engines that were 18.7× slower without it, and a plan/runtime mismatch that silently allocated 6.25 GiB of unbudgeted KV cache

DeepSpeed — distributed training

[Merged pull requests](#)

fixed a ZeRO-3 rollout deadlock caused by unsynchronized generation stopping, and blocked partial Hybrid Engine policy injection for unsupported architectures (validated on H200 and MI250 GPUs)

Triton — GPU compiler

[Merged pull requests](#)

stopped nested-loop fusion from trusting an `llvm.assume` outside the loop, which removed a zero-trip guard and could cause incorrect memory writes (MLIR regression test and H200 reproducer)

verl — RL post-training

[Merged pull requests](#)

restored FSDP value-head critic loading after TRL relocated its value-head model classes

LMCache — KV cache management

[Merged pull requests](#)

rejected non-positive allocations that corrupted the free list, and enforced lazy `%-format` logging (ruff G004) so new f-string logging can no longer land in already-migrated files

vLLM-Omni — multimodal serving

[Merged pull requests](#)

rejected a stray audio placeholder that killed the Qwen2.5-Omni thinker engine when audio is read from video, turning an engine crash into an upfront validation error

Apache TVM — deep-learning compiler

[Merged pull requests](#)

fixed the ONNX Hardmax lowering when the reduced axis has a symbolic extent, where `one_hot` requires a static depth but the importer passed a symbolic variable

JAX — numerical framework

[Merged pull requests](#)

made `jax.scipy.linalg.lu` accept the batched inputs its documentation promises, where only 2-D inputs worked because the post-processing unpacked the shape directly (PR #40931)

FlashAttention — attention kernels

[Merged pull requests](#)

fit the SM90 CuTe forward tile to the block-sparse block size so every sparse layout is accepted, and skipped the causal mask on unmasked KV blocks, cutting the unmasked inner loop from 580 to 477 instructions on H200

Publications

(*Co-first author)

1. Runguo Li. “Spend Bytes on Breadth: Precision–Count Trade-offs for Decode-Time KV Compression in Long Chain-of-Thought Reasoning.” *arXiv preprint arXiv:2610.05685*, 2026.
Single-author work on KV-cache compression during long chain-of-thought decoding: rather than only choosing which tokens to evict, BreadthKV splits a fixed byte budget between token count and precision, calibrating bit-width end to end per model and budget. It beats eviction alone in 17 of 18 settings across three reasoning models and four benchmarks, and at the tightest budget cuts Qwen3-8B AIME samples that hit the length cap from 91% to 40%.
[PDF](#) | [arXiv page](#)
2. Zhi Yang*, Runguo Li*, Qiqi Qiang, Jiashun Wang, Fangqi Lou, Mengping Li, Dongpo Cheng, et al. “FinVault:

Benchmarking Financial Agent Safety in Execution-Grounded Environments. *arXiv preprint arXiv:2601.07853*, 2026.

The first execution-grounded safety benchmark for financial agents: 31 regulatory sandbox scenarios with writable state, 107 real-world vulnerabilities, and 963 test cases. Current defenses do not transfer — attack success rates reach 50.0% and stay non-negligible (6.7%) even for the most robust model.

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3. Yutao Hou*, Runguo Li* et al. “Reasoning-Visual Critical Token Fine-Tuning for Multimodal Reasoning.” Manuscript under review at AAAI 2027.

RVCFT selects which chain-of-thought tokens receive direct supervision using reasoning-relevance and visual-sensitivity signals while retaining the full response as context; at 50% token retention it gives the best average among the evaluated multimodal reasoning benchmarks.

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4. Yeheng Chen*, Runguo Li* et al. “VeriBRT: Plan-Guided, Evidence-Based Automated Bug Reproduction.” Manuscript under review at ICSE 2027.

Represents issue-report behavior as a persistent Two-Axis Plan (a Checklist for where a test must reach, an Intentlist for what it must verify) with reliability-aware evidence and localization coverage as a validity criterion; reproduces 341 of 433 issues on SWT-Bench Verified, 9.7 points above the strongest same-model baseline.

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Research Experience

University of Illinois Urbana-Champaign (UIUC)

Aug 2026–Present

Research Intern, advised by Prof. Minjia Zhang

Urbana-Champaign, IL, USA

- Research on efficient machine learning systems for large models: training and inference efficiency, model compression and sparsity, kernel- and hardware-aware optimization, and system support for LLM agents.
- Bring an upstream-first practice to this work: reproduce failures in production inference and training stacks, then contribute fixes with regression tests and measured before/after numbers.

Shanghai Jiao Tong University

Mar 2026–Jul 2026

Research Intern, advised by Prof. Xiaodong Gu

Shanghai, China

- Researched LLMs for code, with an emphasis on repository-level program understanding, automated bug reproduction, and coding-agent planning and execution.
- Developed VeriBRT, a plan-guided framework that represents issue-report behavior with a persistent Two-Axis Plan and uses reliability-aware evidence for bug-reproducing test generation; set up the evaluation harness on SWT-Bench Verified.

Tencent Youtu AI Lab

Oct 2025–May 2026

Research Intern

Shanghai, China

- Researched LLM-driven adversarial data synthesis and low-resource, multi-label content-safety evaluation, and contributed to a unified multilingual moderation model on an XLM-RoBERTa backbone.
- Developed RVCFT, a selective-supervision method that identifies reasoning-relevant and visually sensitive tokens for multimodal chain-of-thought fine-tuning.

Technical Skills

- **ML Systems and Inference:** MoE expert streaming across multiple drives (0_DIRECT reads, mirrored and sharded layouts), KV-cache management, decode-time KV compression (quantization plus eviction) and quantization internals (int4 per-token-head, Hadamard rotation), memory budgeting, block-sparse and long-context attention, serving engines (vLLM, vLLM-Omni, SGLang, TensorRT-LLM), GPU compiler passes (Triton, MLIR) and kernel-level debugging, and performance profiling.
- **Training and Post-Training Infrastructure:** distributed data/tensor/pipeline parallelism, ZeRO-3 and FSDP, Megatron-LM, DeepSpeed Hybrid Engine, collective synchronization and deadlock debugging, RL rollout systems (verl, TRL), SFT and preference optimization (DPO/GRPO/PPO), distillation, and LoRA/PEFT.
- **Languages and Tooling:** Python, C, CUDA/Triton, SQL, Bash, $\LaTeX$; PyTorch, JAX, Transformers, FlashAttention, FAISS; Git-based upstream contribution, regression testing, reproducible benchmarking, Linux, and Docker.
- **Systems for Agents:** agentic planning and tool use, RAG and long-term memory, execution-grounded evaluation, runtime safety gates, and multi-channel gateways (FastAPI).

Honors & Awards

National Encouragement Scholarship | People’s Scholarship | Shanghai Municipal Scholarship | Social Practice Scholarship